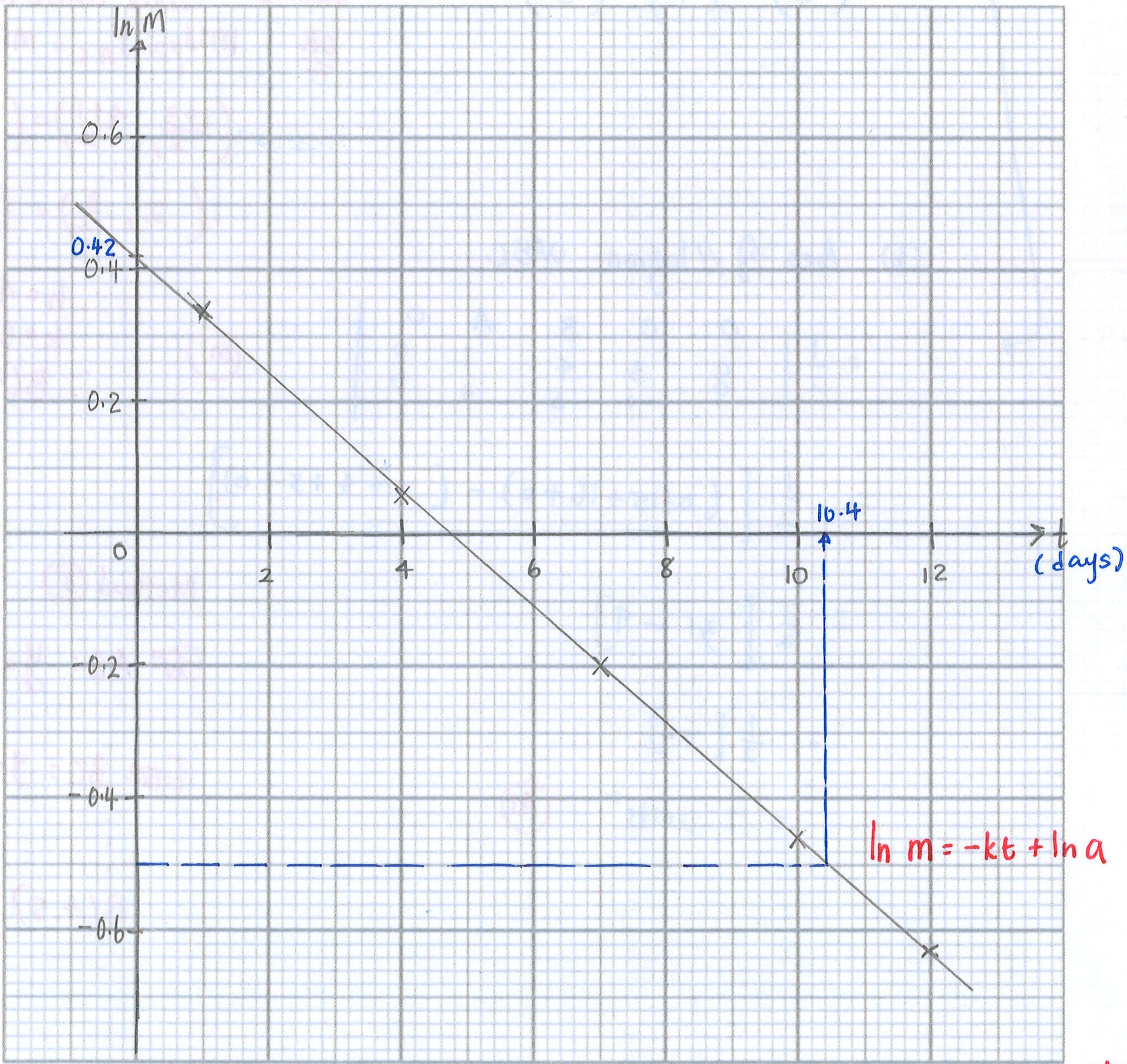
**12** A radioactive substance undergoes decay. Its mass, $M$ grams, after $t$ days is defined by $M = ae^{-kt}$, where $a$ and $k$ are constants.

The table below shows some measurements of $M$ and $t$.

| $t$ (days) | 1 | 4 | 7 | 10 | 12 |
|---|---|---|---|---|---|
| $M$ (grams) | 1.40 | 1.06 | 0.82 | 0.63 | 0.53 |
| $\ln M$ | 0.34 | 0.06 | −0.20 | −0.46 | −0.63 |

**(a)** Plot $\ln M$ against $t$ and draw a straight line graph to illustrate the information. [2]




$M = ae^{-kt}$

$\ln m = \ln ae^{-kt}$

$= \ln a + \ln e^{-kt}$

$\ln m = -kt + \ln a$

$Y = mX + C$

where $Y = \ln m$ , $m = -k$

$X = t$ $\quad C = \ln a$

M1 • Correct points / straight line

m1 • Label / correct scale

[Turn over

**(b)** Use your graph to estimate the value of $a$ and of $k$. [4]

From the graph, $\ln a = 0.42$ M1

$\log_e a = 0.42$

$a = e^{0.42}$

$= 1.52196$

$= 1.52$ (3 sf) A1

gradient $= \dfrac{-0.63 + 0.2}{12 - 7}$ M1

$= -0.086$

$-k = -0.086$

$\therefore k = 0.086$ A1

**(c)** Use your graph to estimate the time taken, to the nearest day, when at least 60% of the radioactive substance has decayed. [2]

60% decayed — 40% present

$40\%\ m = \dfrac{40}{100} \times 1.52196$

$= 0.6087$

$\ln 0.6087 = -0.49629$

$\approx -0.50$ M1

From the graph

$t = 10.4$ days

$\therefore$ time taken $= 11$ days A1

(nearest day)